

Citation Documents / Supporting References

1. Noise Types (Synthetic SEM Dataset)

1.1 Shot + Read Noise

Timischl, Date & Nemoto (2012), "A statistical model of signal-noise in scanning electron microscopy," Scanning 34, 137–144.

<https://pubmed.ncbi.nlm.nih.gov/21898458/>

1.2 PSF / Astigmatism Blur

"Point-Spread Function of the Optics in Scanning Electron Microscopes," arXiv:2407.01439 (2024).

<https://arxiv.org/abs/2407.01439>

1.3 Charging Streaks

Cazaux, J., "Charging effects in the scanning electron microscope," J. Phys. D: Appl. Phys.

<https://iopscience.iop.org/article/10.1088/0022-3735/4/9/002>

Velazco et al. (2025), "Reduction of SEM charging artefacts in native cryogenic biological samples," Nature Communications 16, 5204.

<https://www.nature.com/articles/s41467-025-60545-3>

1.4 Raster Drift + Row Jitter

Cizmar, Vladár & Postek, "Advances in Modeling of Scanning Charged-Particle-Microscopy Images," arXiv:1006.0171 (2010).

<https://arxiv.org/abs/1006.0171>

1.5 Barrel / Pincushion Distortion

Tsuno, K., "Electron lens equipped with three magnetic pole pieces," US Patent 4,412,132 (1983).

<https://patents.google.com/patent/US4412132>

1.6 Pattern Collapse

Noga, Lawson et al. (2009), "Understanding pattern collapse in high-resolution lithography," Proc. SPIE 7273, 727334.

<https://www.spiedigitallibrary.org/conference-proceedings-of-spie/7273/727334/Understanding-pattern-collapse-in-high-resolution-lithography--impact-of/10.1117/12.829142.short>

1.7 Device Geometry

Intel Corporation, "Intel Reinvents Transistors Using New 3-D Structure" (2011).

<https://www.intc.com/news-events/press-releases/detail/655/intel-reinvents-transistors-using-new-3-d-structure>

Hu, Chenming. Modern Semiconductor Devices for Integrated Circuits (Prentice Hall, 2010).

2. Augmentation / Matching Pipeline Citations

2.1 Candidate Generation

Lewis, J.P., “Fast Normalized Cross-Correlation” (1995).

<https://scribblethink.org/Work/nvisionInterface/nip.pdf>

2.2 Two-Stage Retrieval + Rerank

Karpukhin et al., “Dense Passage Retrieval for Open-Domain Question Answering,” EMNLP (2020).

<https://arxiv.org/abs/2004.04906>

2.3 Anti-Aliased Downsampling

Gonzalez, R.C. & Woods, R.E. Digital Image Processing (Pearson) — official book site.

<https://www.imageprocessingplace.com/>

2.4 Held-Out Train/Test Split

Goodfellow, Bengio & Courville. Deep Learning (MIT Press, 2016), Ch. 5.

<https://www.deeplearningbook.org/>